Block-wise Feature Selection in K-Nearest Neighbors for Regression

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Introduction

K-Nearest Neighbor (K-NN) Regression is a robust machine learning algorithm that estimates the regression function using local information. Whereas K-Nearest Neighbors for classification determines the classification based on the number of the K nearest data points and which label appears the most, K-NN for regression is calculated with the average. Variation exists naturally due to many various factors in practical applications (Song et al.). Therefore, feature selection is an important step in reducing computational complexity while retaining essential information. An existing algorithm we considered was RReliefF (Robnik Šikonja & Kononenko), which provides a ranking for each variable while lacking a determination of which features are non-essential. Our method provides a wholistic representation of the effect of disjoint blocks of features removed across a set range of K-values.

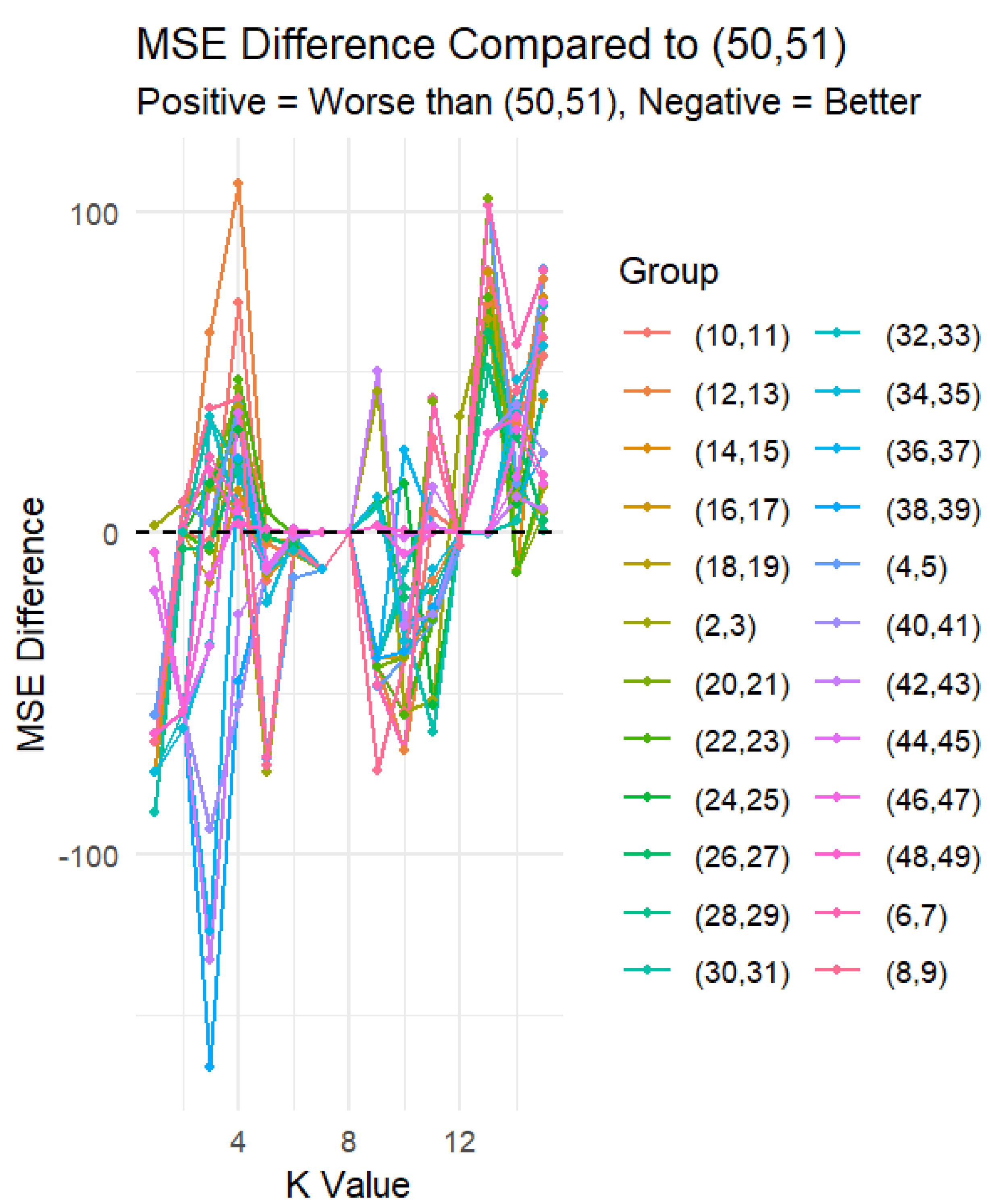
References

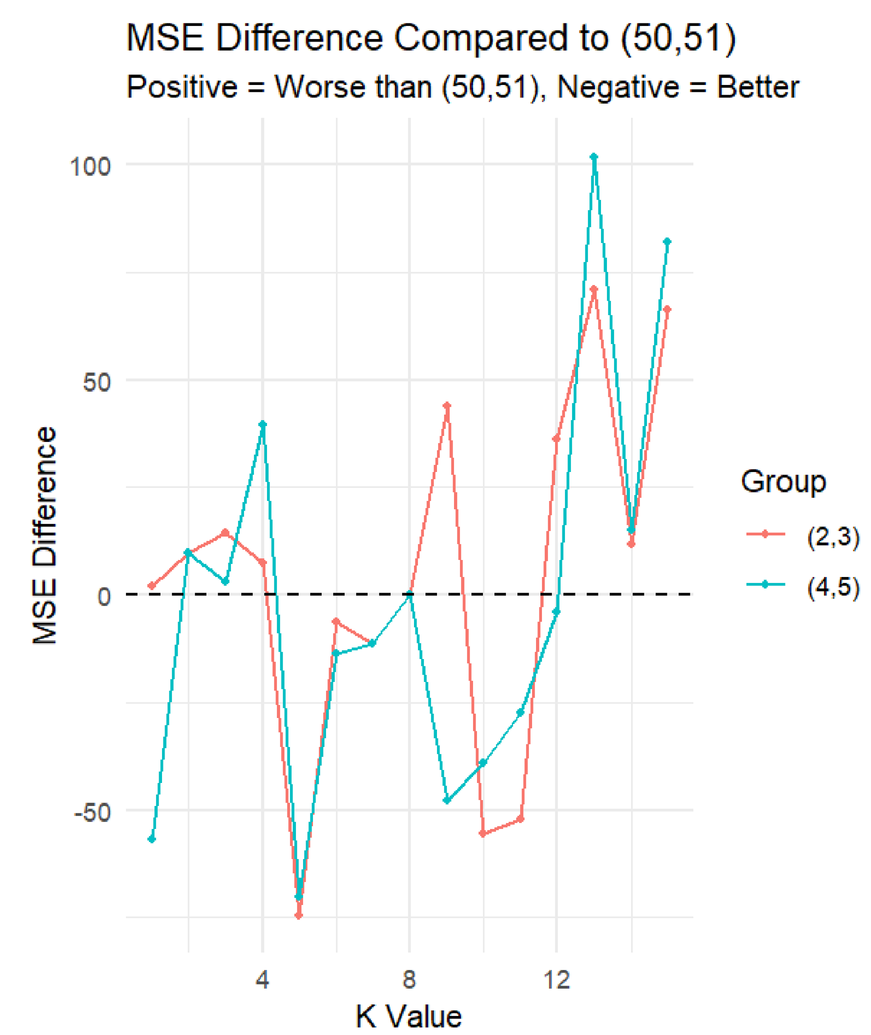
Robnik-Sikonja, M., and I. Kononenko. (1997). "An adaptation of Relief for attribute estimation in regression."

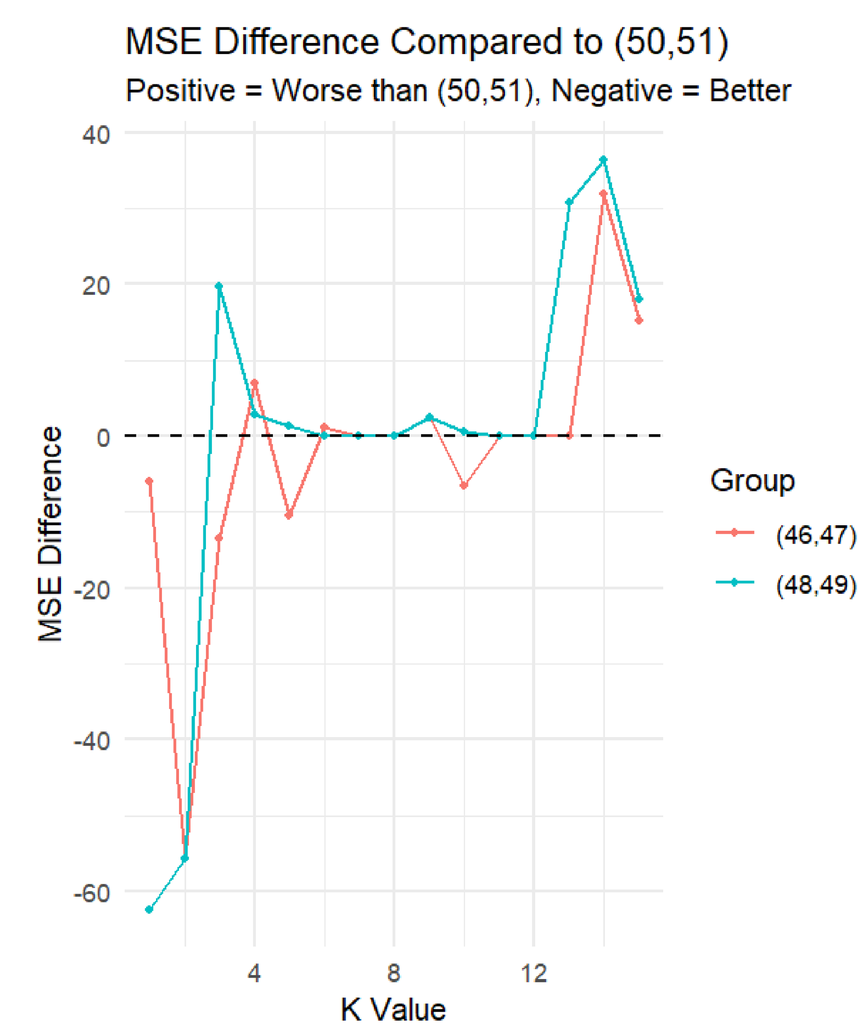
Yunsheng Song, Jiye Liang, Jing Lu, Xingwang Zhao. (2017). "An efficient instance selection algorithm for k nearest neighbor regression." Neurocomputing, Volume 251, Pages 26-34, ISSN 0925-2312.

Method

Our method includes training the K-NN model using disjoint blocks of data where few features are removed for each block. We order features based on their mutual information with the output feature then separate into approximately equal sized blocks. Blocks must be of similar size to retain the complexity of the K-NN model between the predetermined number of features per each block removed. We used a range of 1 through 20 for values of K.







Results

As small values of K risk overfitting while large values of K risk underfitting, the trends we see on both ends are less significant than the pattern emerging in the central values of K. Input features with higher mutual information with the output feature had a greater difference in error curves than those with lower mutual information. Data was simulated with noise, so variation in differences is expected. The total number of features is 51 with 50 input features and the total number of instances is 100.

Conclusion

When only a certain number of features can be included in the training for the K-NN model, RReliefF is a simple algorithm that shall suffice. However, when needing to analyze the effects of removal of input features from a more wholistic perspective, block-wise feature selection utilizing the whole error curve is a potential option. We hypothesize that error curves of feature blocks with less significant differences between those of the highest mutual information shall remain. Error curves of feature blocks with more significant differences than the feature blocks shall be considered for elimination.

As K-NN models are computationally expensive to train, this method of feature selection is similarly computationally expensive. In future work, we will take into consideration of the effect of the block sizes and the range of values of K.